

Reconstructing a radio map from 1–20% of it

Classical interpolation baselines on a Sionna RT street canyon, scored two ways: filling scattered gaps, and predicting into a block of street that was never measured. The two answers differ by a factor of five.

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Hosted by Prof John Thompson, IDCoM, University of Edinburgh 18 September 2026

2.08 dB best RMSE at 20% random sampling	9.68 dB best RMSE at 20% into a 60 m hold-out block	2.3 dB all that 20× more data buys under hold-out
0.23 dB uncertainty of the ground truth itself		

01

What was asked, and what came back

The task was to generate a Sionna RT radio map, keep a small percentage of the spatial samples, rebuild the full map with IDW, RBF and kriging, and compare RMSE and MAE across sampling densities, with a spatially held-out region rather than random sampling alone.

That is done, at 3.5 GHz on the `simple_street_canyon` scene, and this note reports it. Three things came out of it that were not obvious at the start.

- **The ground truth is noisier than the thing being measured**, at the ray counts published work uses. Fixing that costs 86 seconds of compute, so it seemed worth fixing before anything else.

- **The contiguous hold-out is not a harder version of random sampling. It is a different problem.** Errors are about five times larger, and they stop improving with more data.
- **Under hold-out the method ranking becomes unstable**, and which stretch of street you delete matters far more than which interpolator you use.

Status

Everything here is a single scene, a single transmitter and a single frequency. It is a starting point, not a result about radio maps in general. Section 6 lists what would have to change before any of it generalises.

02

Setup

Sionna RT 2.0.1, the built-in `simple_street_canyon` scene, one isotropic transmitter on a rooftop at 30 dBm. Line of sight, specular reflection and diffraction on; diffuse scattering and refraction off. The measurement plane is at 1.5 m with 1 m cells, giving a 122×187 grid.

Of the 22,814 cells, 5,236 sit under building geometry and are excluded, found by casting a vertical ray from each cell centre. A further handful of open-street cells are never reached by any ray. That leaves **17,547 valid cells**, 76.9% of the grid, spanning -150 to -73 dB with a standard deviation of 17.6 dB.

Two things the docs do not say

`path_gain` is returned as a *linear* power ratio, not dB. Values here are of order 10^{-9} .

`refraction` defaults to on. Left on, rays leak into building interiors and those cells stop being structurally unreachable, which quietly changes what an accessibility mask means.

scene	simple_street_canyon	carrier	3.5 GHz
cell size	1.0 m	grid	122×187
tx position	(-33, 11, 32) m	tx power	30 dBm
max depth	3	rays per tx	2×10^8
valid cells	17,547	solver time	86 s, 2 CPU cores

How much of a radio map is simulator noise

Before comparing reconstruction errors of a few dB, it is worth asking how repeatable the ground truth is. Tracing the same scene twice with different random seeds gives two maps that should be identical and are not. The RMSE between them is a direct measure of how much of the map is Monte Carlo noise; a single map sits that distance divided by $\sqrt{2}$ from the converged mean.

At **2×10^5 rays** — the count used by the closest published Sionna RT reconstruction study — 29% of open-street cells receive no ray at all, and the cells that do receive rays disagree between seeds by 4.75 dB. Reconstruction errors reported in this literature are typically 2–5 dB. At that ray count the simulator disagrees with itself by more than the effect being measured.

From 10^6 rays upward the error falls as $N^{-1/2}$, as it should for a Monte Carlo estimator. This study runs at **2×10^8** , where a single map sits 0.23 dB from the converged mean, roughly an order of magnitude below the smallest reconstruction error reported below. That costs 86 seconds.

A second check on the same point

The fitted variogram has a nugget of essentially zero. A noisy ground truth would show a large one, because independent per-cell noise appears as variance that does not vanish at zero lag. The two diagnostics agree.

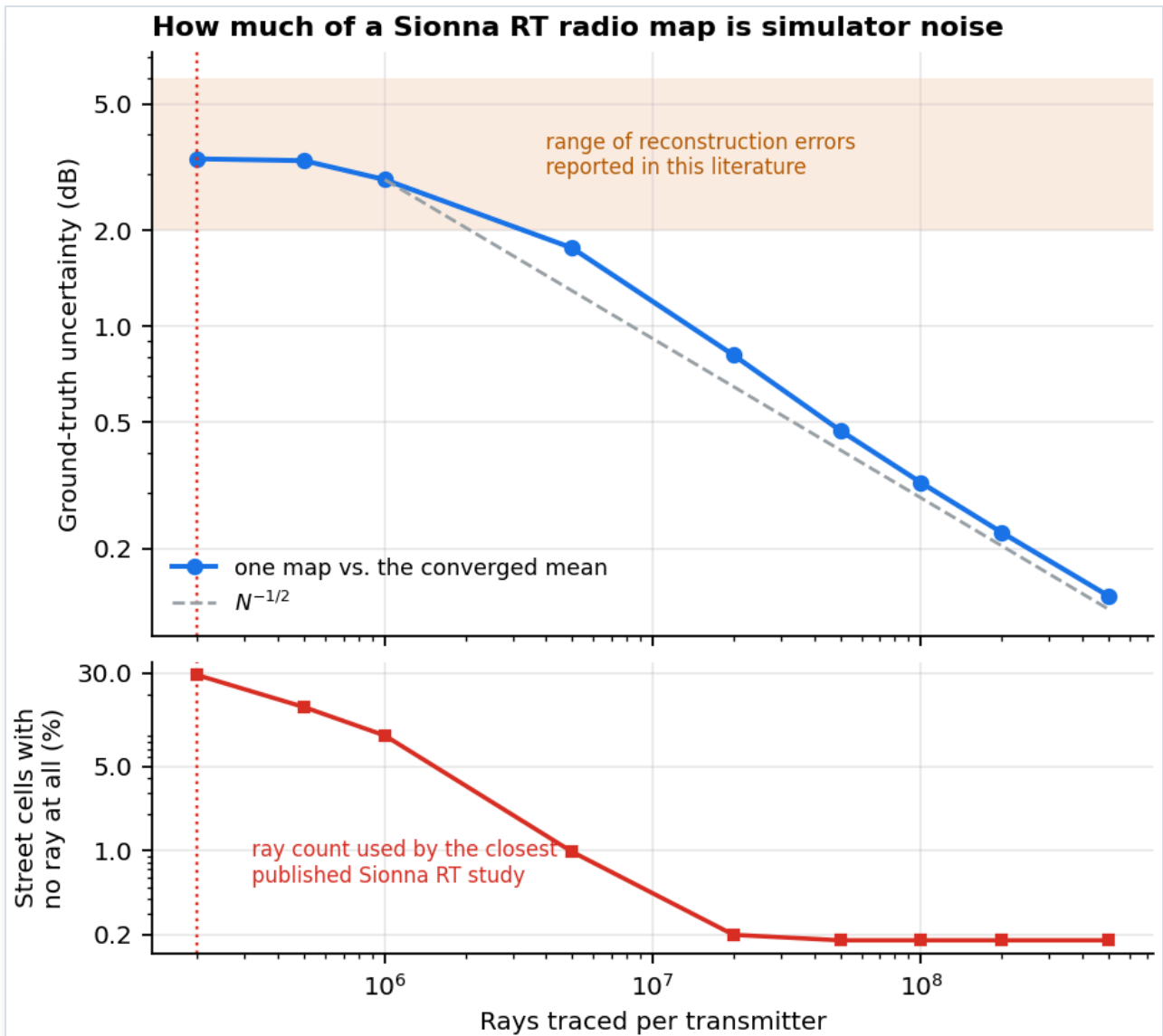


FIG 1 The ray count sets a floor on what any reconstruction study can resolve. Below roughly 10^7 rays that floor sits inside the range of the errors being reported.

Two independent traces of the identical scene, compared over open-street cells reached in both.

RAYS PER TX	STREET CELLS WITH NO RAY	SEED-T0-SEED RMSE	ONE MAP VS. MEAN	SOLVE TIME
2×10^5	28.94%	4.75 dB	3.36 dB	<1 s
1×10^6	9.10%	4.08 dB	2.89 dB	1 s
5×10^6	0.98%	2.49 dB	1.76 dB	3 s
2×10^7	0.20%	1.15 dB	0.81 dB	9 s
5×10^7	0.18%	0.67 dB	0.47 dB	22 s
1×10^8	0.18%	0.46 dB	0.32 dB	44 s
2×10^8	0.18%	0.32 dB	0.23 dB	86 s
5×10^8	0.18%	0.20 dB	0.14 dB	269 s

04

Random sampling and contiguous hold-out

Two sampling designs, same map, same methods.

Random draws observations uniformly from the valid cells and scores on the rest. Every unobserved cell has an observation nearby, so this measures gap filling. It is what nearly all of this literature does.

Block removes a contiguous 60 m square entirely, draws observations from what remains, and scores only inside the square. Nothing inside is ever observed, so it measures extrapolation into unseen ground. The block side is set to about twice the 27.8 m correlation range fitted from the variogram: a block narrower than the correlation range is still measuring interpolation.

Six block positions are swept, three sampling seeds each, so block placement is a replicate rather than a single choice. Hyperparameters for every method are chosen by 5-fold cross-validation on the observations only, never against the held-out truth.

Why score only unobserved cells

Including observed cells would flatter IDW, RBF and kriging, all of which reproduce their own observations exactly, and the flattery would grow with sampling density. Papers differ on this; one

that formalises it finds the choice can change the spread between best and worst method by a factor of five.

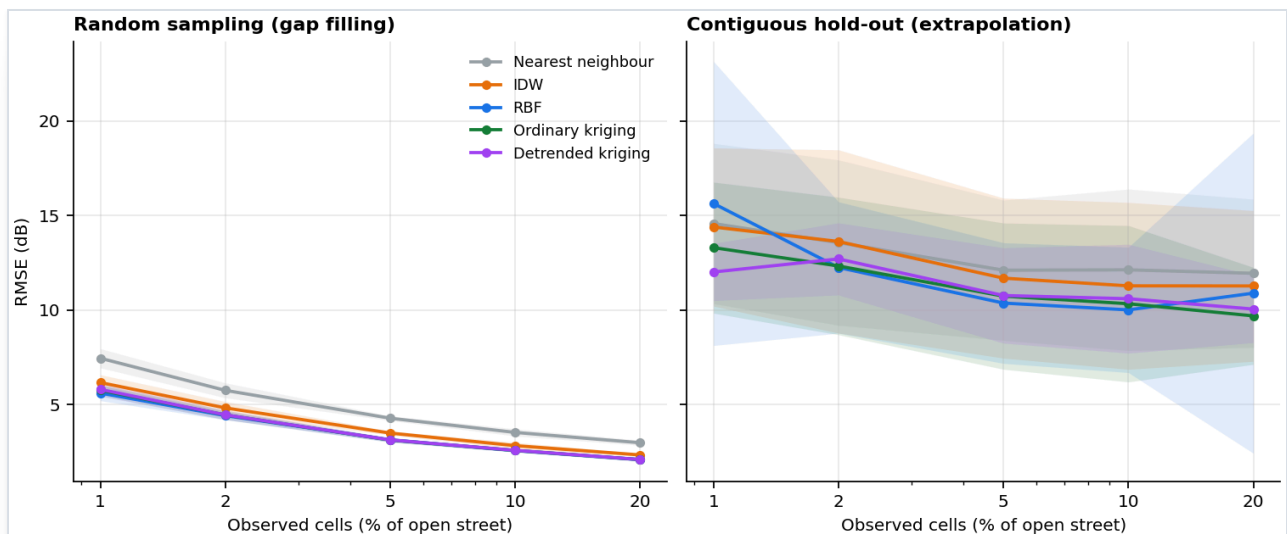


FIG 2 Same map, same methods, two designs. Shaded bands are one standard deviation: across seeds on the left, across seeds and six block positions on the right.

Under random sampling, going from 1% to 20% of the map cuts the error by 63%. Under contiguous hold-out the same twentyfold increase in data buys **2.3 dB**, and the curve is flat from 5% onward. Samples outside a gap do not help you fill it once you have enough of them to pin down the surroundings.

RMSE in dB, mean $\pm$ standard deviation. Random: 5 seeds. Block: 6 block positions $\times$ 3 seeds = 18 runs per cell.

METHOD	1%	2%	5%	10%	20%
RANDOM SAMPLING					
Nearest neighbour	7.44 ± 0.50	5.74 ± 0.39	4.27 ± 0.12	3.51 ± 0.19	2.97 ± 0.13
IDW	6.15 ± 0.41	4.81 ± 0.33	3.48 ± 0.10	2.82 ± 0.17	2.32 ± 0.08
RBF	5.60 ± 0.40	4.40 ± 0.21	3.12 ± 0.12	2.56 ± 0.08	2.08 ± 0.10
Ordinary kriging	5.75 ± 0.30	4.44 ± 0.25	3.11 ± 0.10	2.56 ± 0.08	2.08 ± 0.08
Detrended kriging	5.79 ± 0.36	4.45 ± 0.25	3.12 ± 0.10	2.57 ± 0.08	2.08 ± 0.08
CONTIGUOUS 60 M HOLD-OUT					
Nearest neighbour	14.57 ± 4.24	13.56 ± 4.37	12.10 ± 3.71	12.12 ± 4.27	11.94 ± 3.92
IDW	14.39 ± 4.18	13.62 ± 4.85	11.68 ± 4.23	11.27 ± 4.41	11.27 ± 3.99
RBF	15.63 ± 7.51	12.24 ± 3.47	10.36 ± 3.19	10.00 ± 3.32	10.88 ± 8.47
Ordinary kriging	13.29 ± 3.46	12.32 ± 3.64	10.73 ± 3.87	10.33 ± 4.14	9.68 ± 2.56
Detrended kriging	12.01 ± 1.51	12.70 ± 1.91	10.76 ± 2.52	10.59 ± 2.88	10.04 ± 1.77

WHAT THE RANKING DOES

Under random sampling the ordering is stable and boring: RBF, ordinary kriging and detrended kriging are indistinguishable, IDW trails by about 0.24 dB, nearest neighbour by 0.9 dB. Standard deviations across seeds are 0.08–0.5 dB, so the ordering is real.

Under hold-out the ordering stops being stable. Standard deviations are 1.5–8.5 dB, which is larger than the gaps between methods, so most of the ranking is not resolved by this data. Two things do survive:

- **RBF is the least reliable extrapolator.** It has the best mean at 5% and 10% and the worst variance at 1% and 20% (± 8.47 dB). A thin-plate spline has nothing holding it down outside the convex hull of its data.
- **Detrending buys stability, not accuracy.** Fitting a log-distance path-loss trend before kriging the residual gives roughly the same mean error but halves to quarters the variance across block positions. At 1% it is both the best and by far the steadiest. The trend fit, $\text{gain}_{\text{dB}} = -28.6 - 40.9 \log_{10}(d)$, takes the residual standard deviation from 17.7 to 14.5 dB and the correlation range from 27.8 m to 16.2 m.

The honest headline is that *where* the gap is matters more than which interpolator fills it.

05

Why the hold-out is so much harder

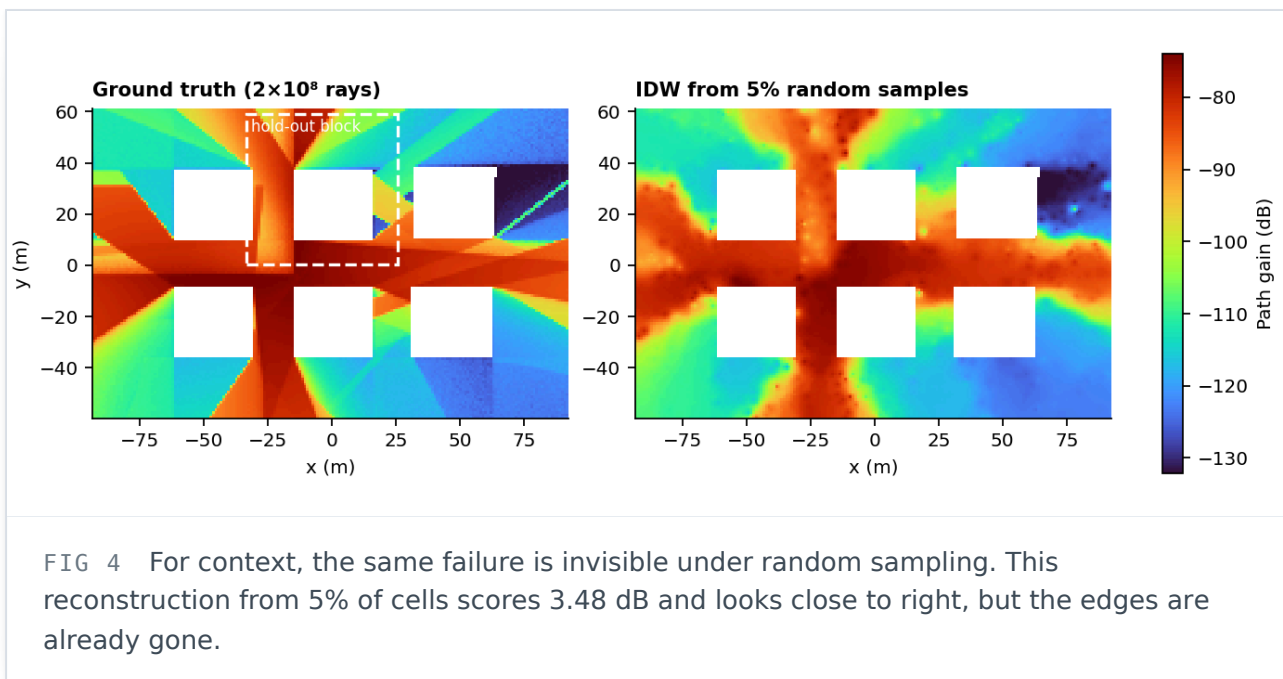
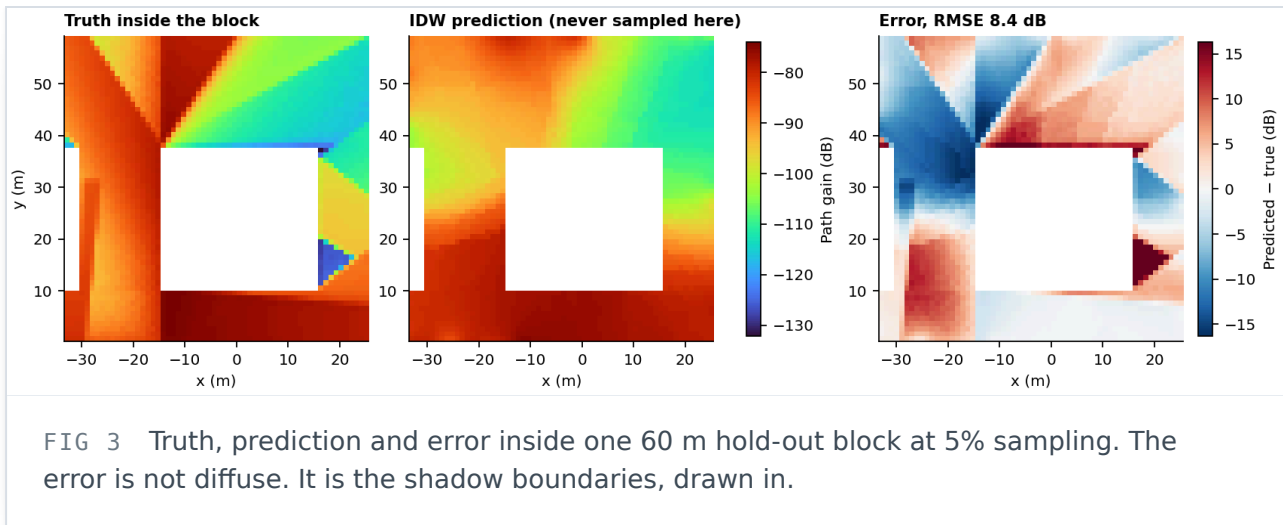
Looking at one block explains the numbers. The truth inside it is not a smooth field. It is piecewise smooth, cut by sharp edges where a diffraction wedge or a specular reflection boundary starts and stops. Those edges are the geometry of the canyon written into the map.

IDW, RBF and kriging all assume the field varies smoothly with distance. Under random sampling that assumption survives, because observations land on both sides of every edge and the edge gets pinned down by data. Under hold-out there is nothing inside the block to pin anything, so every method returns a smooth surface, and the error map is the edges themselves at ± 15 dB.

This is why more samples outside do not help. The missing information is not "what is the level around here", which the surroundings already answer. It is "where does the shadow boundary run", which is a fact about geometry that no amount of sampling elsewhere in the street recovers.

The practical reading

If the aim is a radio map for a region you cannot measure, a purely spatial interpolator is the wrong tool no matter how much data you feed it from elsewhere. Something has to carry the geometry: a building mask, a path-loss model, or a learned prior.



06

What this does not show, and what would come next

One scene, one transmitter position, one frequency, one solver, no measurement noise added on top of the simulation. The block results come from six positions in a single street layout, and the spread across those six is large enough that a different layout could plausibly reorder the methods. Nothing here should be read as a general claim about interpolators.

THREE THINGS WORTH DOING NEXT

- **Does the required sampling density scale with frequency?** Theory says the estimation problem is invariant to scaling wavelength and distance

together, so spacing should track λ and density should go as f^2 . At 28 and 39 GHz the canyon becomes near-binary line of sight and the map develops harder edges, so the smoothness assumption should break down faster than that law predicts. Running the same sweep at 3.5, 28 and 39 GHz on this scene would test it directly. The grid has to be fixed in wavelengths or in metres, deliberately and stated, because the two give different questions.

- **Does a building mask close the gap?** Section 5 says the missing information is geometric. Giving the interpolator the building footprints, which are free in a simulation and cheap in reality, is the smallest change that could test whether that diagnosis is right.
- **dB or linear.** Everything here interpolates in dB, which is the field convention and the domain where the residuals are approximately Gaussian. Part of the literature scores in linear power instead, which weights cells near the transmitter far more heavily. No one appears to have quantified the difference. It is a small ablation to run and it would settle which numbers are comparable to which papers.

Second task

The Sionna RT versus 3GPP 38.901 UMi comparison is still open and would sit naturally alongside the frequency sweep, since both need the same multi-band maps.

07

Reproducing this

The code is a small package plus three scripts, and a self-contained notebook that runs top to bottom on a free Colab CPU runtime. No GPU is required; the 2×10^8 -ray ground-truth map takes 86 seconds on two CPU cores.

Nineteen known-answer checks run in `tests_pipeline.py` and all pass: every interpolator reproduces its own observations; thin-plate splines are exact on a linear field and IDW is not; the variogram fitter recovers a planted correlation range of 15 m as 15.5 m; detrended kriging beats ordinary kriging when extrapolating a pure trend; the sampling designs never leak an observation into a hold-out block; and cross-validation prefers a sensible IDW power over a degenerate one.

DECISIONS A READER MIGHT WANT TO ARGUE WITH

- Errors are scored on unobserved valid cells only.
- Cells under building geometry are excluded, found by vertical ray casting rather than by thresholding the gain.
- Kriging uses a moving neighbourhood of the 32 or 64 closest observations, chosen by cross-validation. Global kriging would need an n^3 solve at these sample counts.
- The variogram is fitted on a 10% draw, which is what a real user would have, not on the full map.

Interim report on the first of two tasks suggested by Tianrun Qi, Southeast University, visiting IDCoM at the University of Edinburgh until November 2026. Prepared by Ali Tagiyev, Year 12, Kings' School Al Barsha, Dubai. Code and figures available on request.